Algorithm S1 | Shared-forecast MPC-DWA cycle

Input: current UAV state, latest 10 obstacle observations, previous MPC solution

for each 0.1-s update:

1. Predict 30 future obstacle positions from the latest 10 observations with LSTM.
2. Share this time-indexed forecast with MPC and DWA.
3. Initialize/shift MPC guide; update trust from safety, progress, tracking and smoothness.
4. Split guided/exploratory DWA samples using trust/MPC guidance; rank by score.
5. Roll out candidates over the same 30-step forecast.
6. Retain clearance >0.25 m, corridor and braking passes; score survivors.
7. Choose the top safe DWA action/envelope; otherwise screen emergency actions.
8. If none passes, stop; otherwise re-screen the selected emergency action.
9. Re-solve MPC on the same forecast: DWA envelope normally; physical bounds in recovery.
10. Apply screened DWA/emergency action for 0.1 s; update state/history and repeat.

Output: checked next action and updated MPC guide, or a fail-closed stop